Contact

danielhardestylewis@gmail.com

www.linkedin.com/in/dhardestylewis (LinkedIn)
dlewis.ai (Portfolio)
github.com/dhardestylewis (Portfolio)
homecastr.com (Portfolio)

Top Skills

Database Administration
Data Analytics
RDBMS

Languages

Spanish (Professional Working)
French (Limited Working)
English (Native or Bilingual)

Honors-Awards

Fellowship

Publications

Artificial Intelligence for Modeling Complex Systems

Daniel Hardesty Lewis

Founder, Worldcastr | prev SPC, DARPA World Modelers, TACC
New York City Metropolitan Area

Summary

building the foundation model for how places change and the platform for high-stakes decisionmakers to interact with itprev helped TACC team go from $110M XSEDE2 to $1.27B LCCF grant, to become leading supercomputing facility in the nationwent from tech lead on $300K NSF IS-GEO award to tech lead on $40M-$70M FEMA & DARPA contractsoperationalized and scaled SOTA models in climate, economics, migration, land use, hydrology, and precipitation alongside the best in their fields, whether public or private sector peak compute spend of $1M overnight on 800K simultaneous nodes to produce millions of earth images at the highest qualitycross registered for credit graduate and doctoral courses across Columbia Law, Statistics, Computer Science, Financial Engineering, Design, Urban Planning, Real Estate, and morecollaborated with luminaries in supercomputing and geomorphology such as Eijkhout & Passalacquabecame co-instructor of graduate coursework at UT Austin within a month of receiving my undergraduate degree in mathematics

Experience

worldcastr
Founder
December 2025 - Present (11 months)
New York, New York, United States

the foundation model of how places change
the platform for high-stakes decisionmaking
worldcastr.com

South Park Commons
Member
June 2026 - September 2026 (4 months)
New York, New York, United States

Summit Geospatial

Founder

November 2023 - August 2025 (1 year 10 months)

the highest quality elevation data in Texas

Texas Advanced Computing Center (TACC)

5 years 7 months

Senior Data Scientist and Technical Lead

August 2021 - August 2023 (2 years 1 month)

Austin, Texas, United States

a tech lead on $40M-$70M FEMA & DARPA contracts: TDIS & DARPA World Modelers MINT

Led a principal project of $40M disaster resiliency initiative, TDIS https://www.glo.texas.gov/disaster-recovery/planning-studies/texas-disaster-information-system

Contributed research towards Paola Passalacqua's 2022 Bagnold Medal, one of the highest honors in geomorphology https://www.egu.eu/awards-medals/ralph-alger-bagnold/2022/paola-passalacqua/

Contributed to an NSF Big Idea https://www.nsf.gov/about/history/big-ideas#navigating-the-new-arctic-3f4

This work ramped up to peak in compute spend at $1M overnight jobs on 800K simultaneous nodes, producing millions of earth images at the highest quality.

I operationalized and scaled SOTA models in climate, economics, migration, land use, hydrology, and precipitation alongside the best in their fields, whether public or private sector.

I design and implement intelligent decision support systems to support computational science, disaster mitigation and response, and resiliency research projects:
- scale climate models on the most powerful computers in the world
- develop relationships with university, state, and federal agencies (NOAA, US Army Corps of Engineers, Texas General Land Office, Texas General Information Office, Texas Water Development Board, TNRIS, USC, UT, A&M, etc)

- prepare computing facility for rapid response to drought, hurricane, flood disasters with replicable scaling of engineering workflows and training

Data Scientist and Research Engineer
February 2018 - July 2021 (3 years 6 months)
Austin, Texas, United States

Deep Learning for Petrobras

DARPA World Modelers: disaster resiliency in East Africa https://www.darpa.mil/research/programs/world-modelers

I contribute research towards integrated modelling and water resource management within the state of Texas and abroad, and I've applied machine and deep learning approaches to problems for the energy industry.

The University of Texas at Austin
Co-instructor
January 2018 - December 2020 (3 years)
Austin, Texas, United States

Machine Learning for the Geosciences
Intelligent Systems for the Geosciences
Scientific Computation

ICC Austin
6 years

System Architect
September 2014 - August 2020 (6 years)
Austin, Texas, United States

Re-designed and implemented entirely new internet infrastructures for 9 buildings serving 115 students, enabling gigabit speeds at each. Re-positioned new enterprise-level wireless access points. Trained organizational capacity to manage new infrastructure.

Conflict mediator
June 2015 - August 2018 (3 years 3 months)
Austin, Texas

Mediated disputes between members and their housing coops. Familiar with all applicable policies and bylaws. Guided facilitations which reached mutually agreed-upon solutions, to help ensure the continued success of this tenant self-managed housing organization.

Technology Officer
September 2014 - December 2017 (3 years 4 months)
Austin, Texas Metropolitan Area

Set up and managed public computers for a 20+ tenant building.

ICPSR at University of Michigan Institute for Social Research
Program Scholar
June 2019 - July 2019 (2 months)
Ann Arbor, Michigan, United States

The ICPSR Summer Program provides rigorous, hands-on training in statistical techniques, research methodologies, and data analysis. ICPSR Summer Program courses emphasize the integration of methodological strategies with the theoretical and practical concerns that arise in research on substantive issues.

Courses enrolled:
- Machine Learning: Applications and Opportunities in the Social Sciences
- Intro to Network Analysis
- Math for Social Science III

Texas Advanced Computing Center (TACC)
Undergraduate Research Assistant
May 2017 - August 2017 (4 months)
Austin, Texas, United States

With a $4000 fellowship from the Texas Institute for Discovery Education in Science, I helped update Texas's groundwater availability models to a more contemporary format

MICEFA
Information Technology Intern
October 2016 - January 2017 (4 months)
Paris, Île-de-France, France

Developed Bash workflow to transform data from a proprietary database into SQL

The University of Texas at Austin
Undergraduate Research Assistant
May 2013 - December 2013 (8 months)
Austin, Texas Metropolitan Area

The boundaries of the stellar evolution code MESA were tested, especially with regard to the high-mass, oxygen/neon-core white dwarf regime. Theoretical

models were run, and the existence (by the standard physics as implemented in the program) of such a regime was found from these calculations.

Education

Columbia University

Master of Science - MS · (2024 - 2026)

Columbia Engineering

Cross-registered · (2024 - 2026)

The University of Texas at Austin

Bachelor of Science - BS, Pure Mathematics · (2012 - 2017)

The University of Texas at Austin

Certificate, Scientific Computation